BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

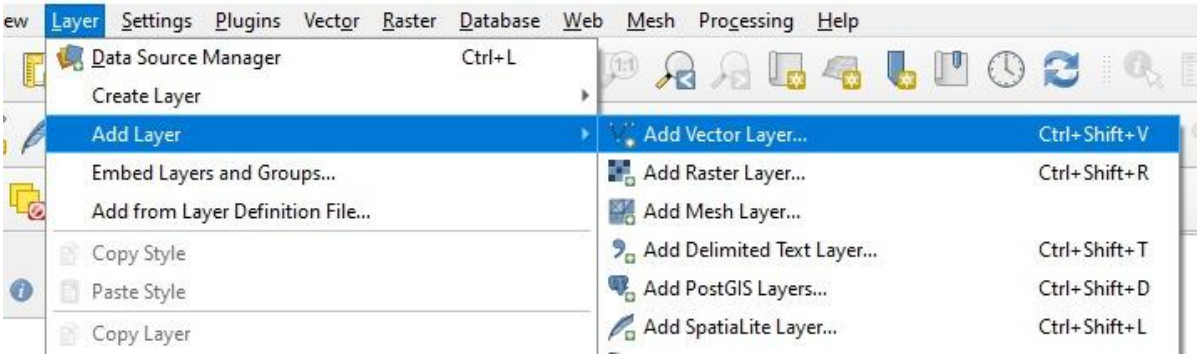
Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

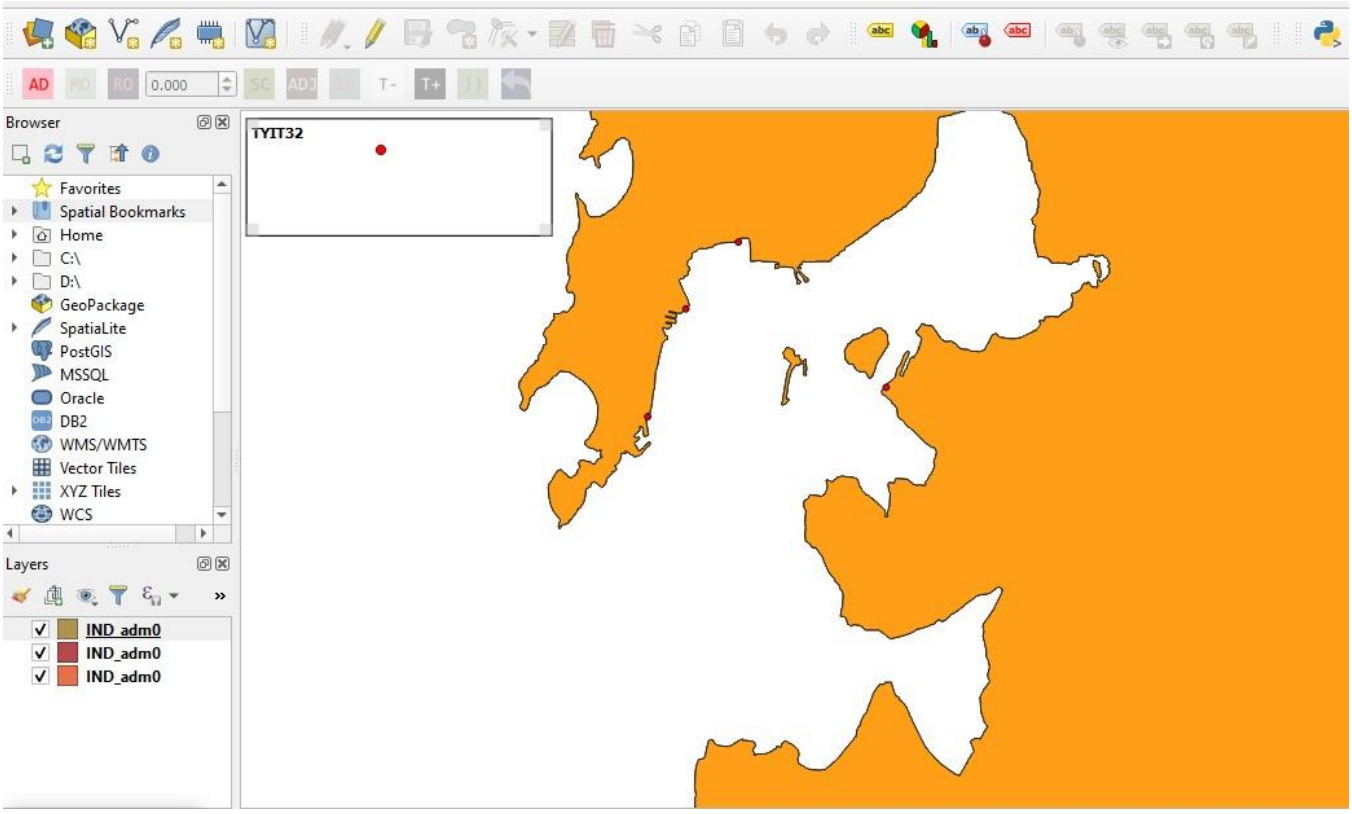
Course Name: Geographic Information system **Page no:**

Practical Number:- 6

A. Georeferencing Topo Sheets and Scanned Maps

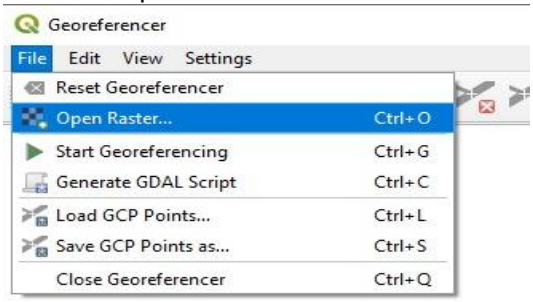
➢ Start a new project ➢ Go to Layers → Add Layer → Add vector Layer ➢ Select GIS_Workshop\Manual\Prac06\IND_adm0.shp ➢ Zoom in to Mumbai region in the layer





➢ Go to Raster → Georefrencer

➢ File → Open Raster



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

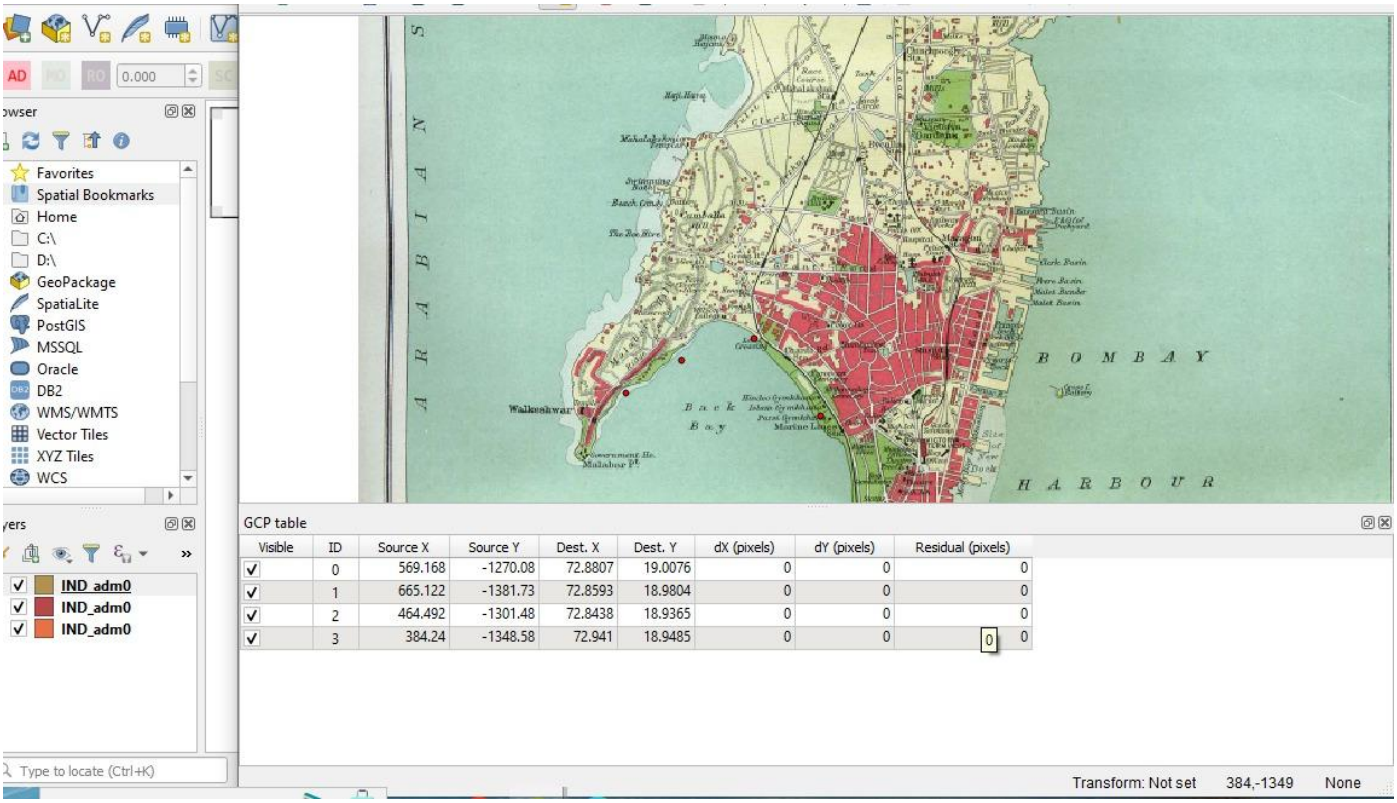
Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

Course Name: Geographic Information system **Page no:**

Practical Number:- 6

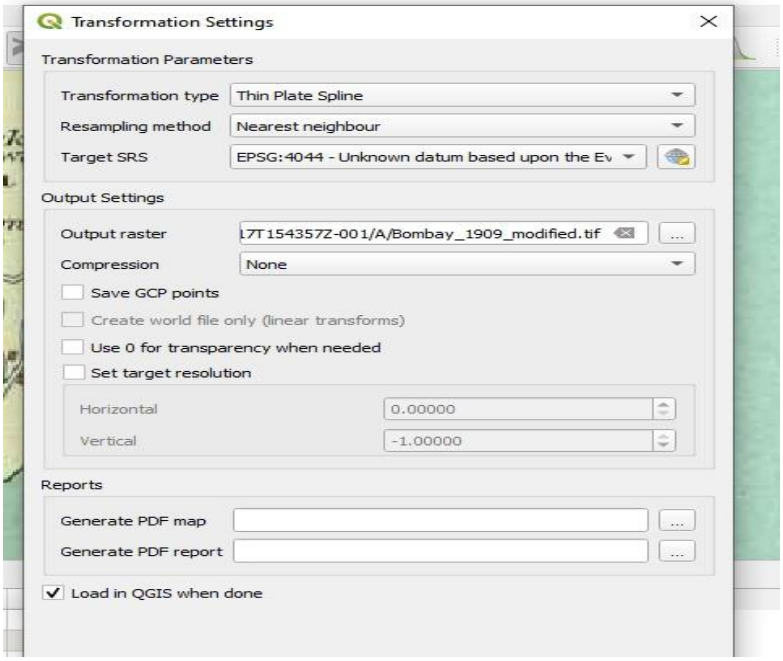
➢ Select file "1870_southern-india_3975_3071_600.jpg" from project data folder

➢ In Georeferencer window Go to Edit → Add Points



➢ Go to Settings →Transformation Settings ▪

Select Transformation type → Thin Plate Spline ▪ Re-sampling Method → Nearest Neighbour ▪ Target TRS → Everest 1830 datum: EPSG 4044 ▪ Select Output Raster Name and Location ▪ Check the Load in QGIS When Done Option ▪ Press "OK"



Teacher's Signature

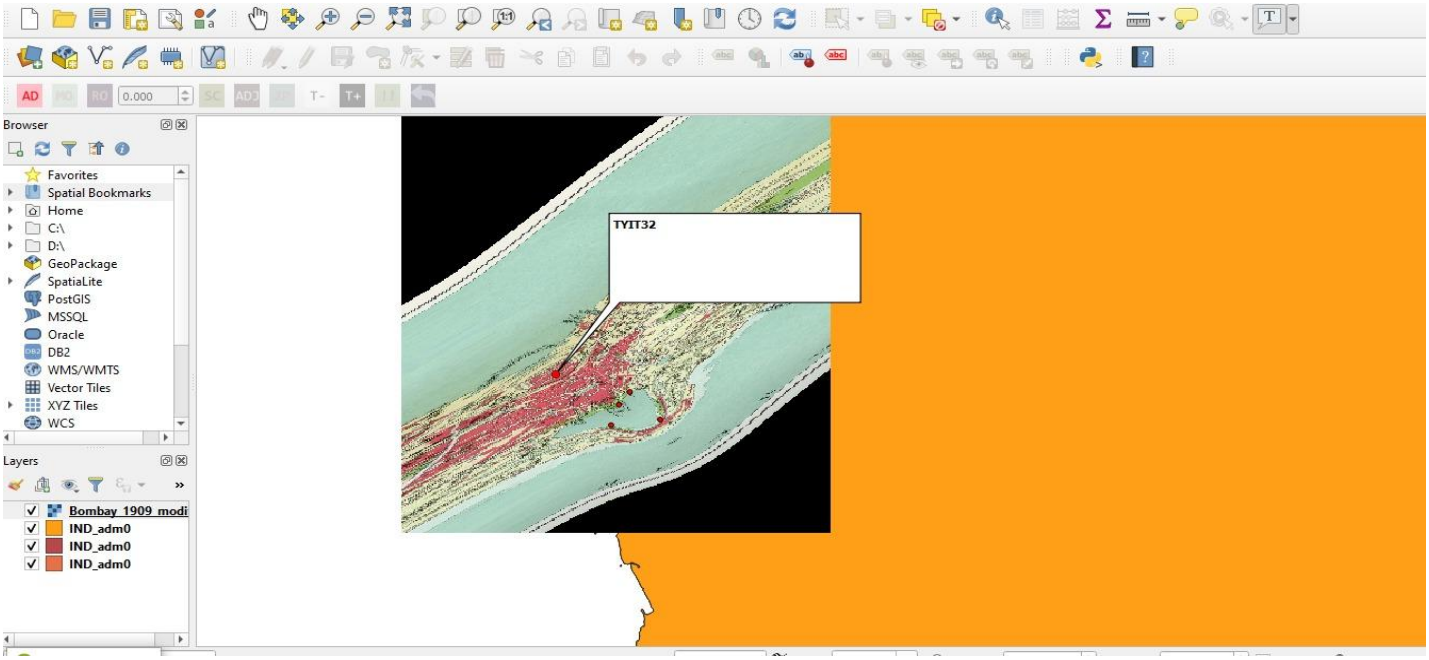
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

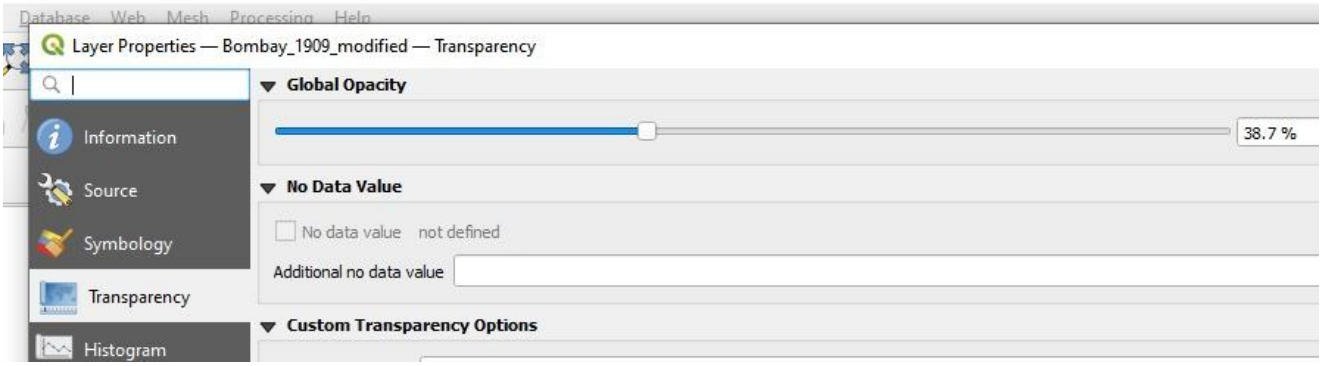
Course Name: Geographic Information system **Page no:**

Practical Number:- 6

➢ Press "RUN"



➢ Set Transparency level of raster layer to appropriate level.



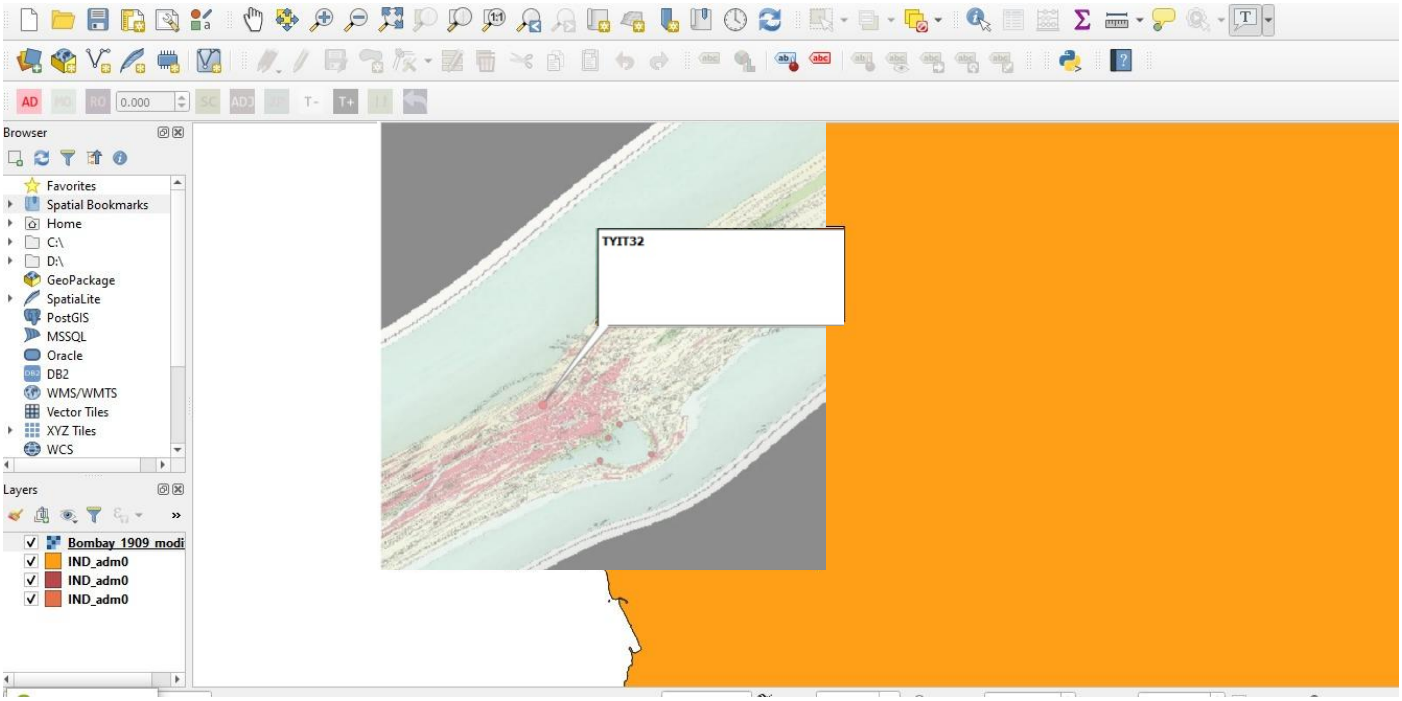
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

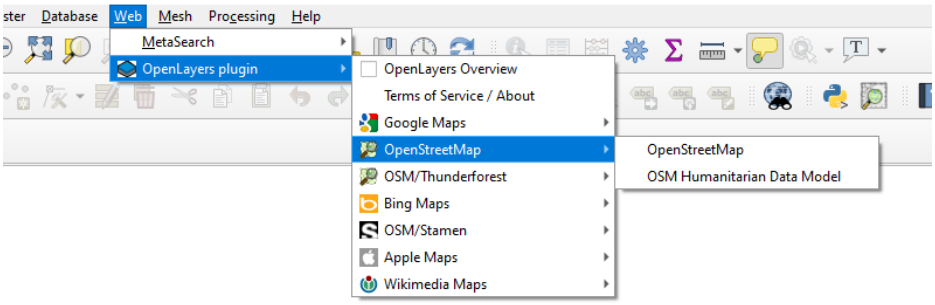
Course Name: Geographic Information system **Page no:**

Practical Number:- 6

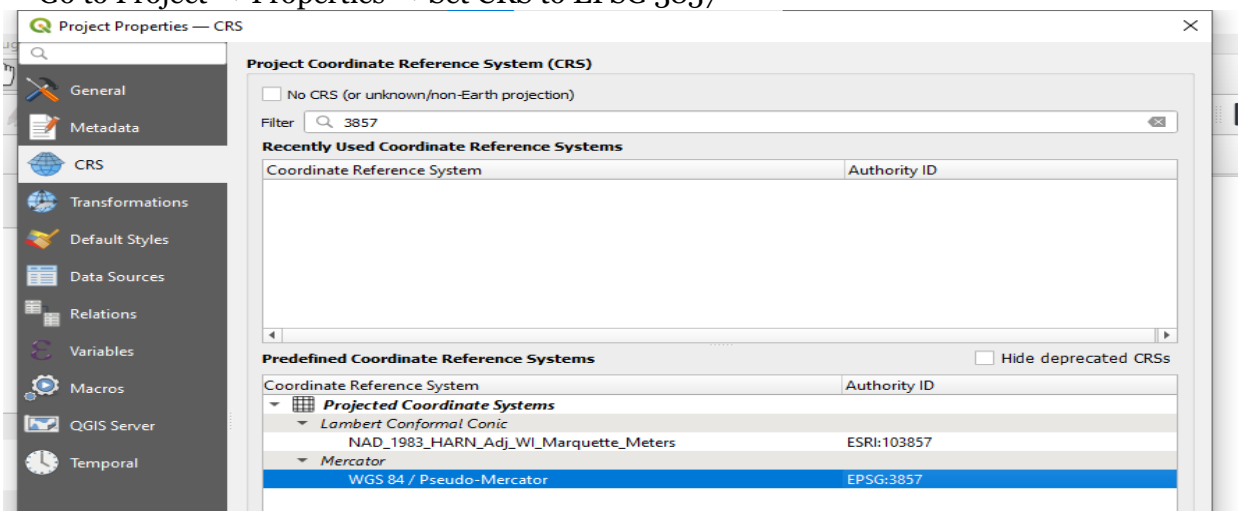


B. Georeferencing Aerial Imagery

➢ Install plugin OpenStreetMap ➢ Go to Web Menu → OpenLayerPlugin → OpenStreetMap→ OpenStreetMap



➢ Go to Project → Properties → Set CRS to EPSG 3857



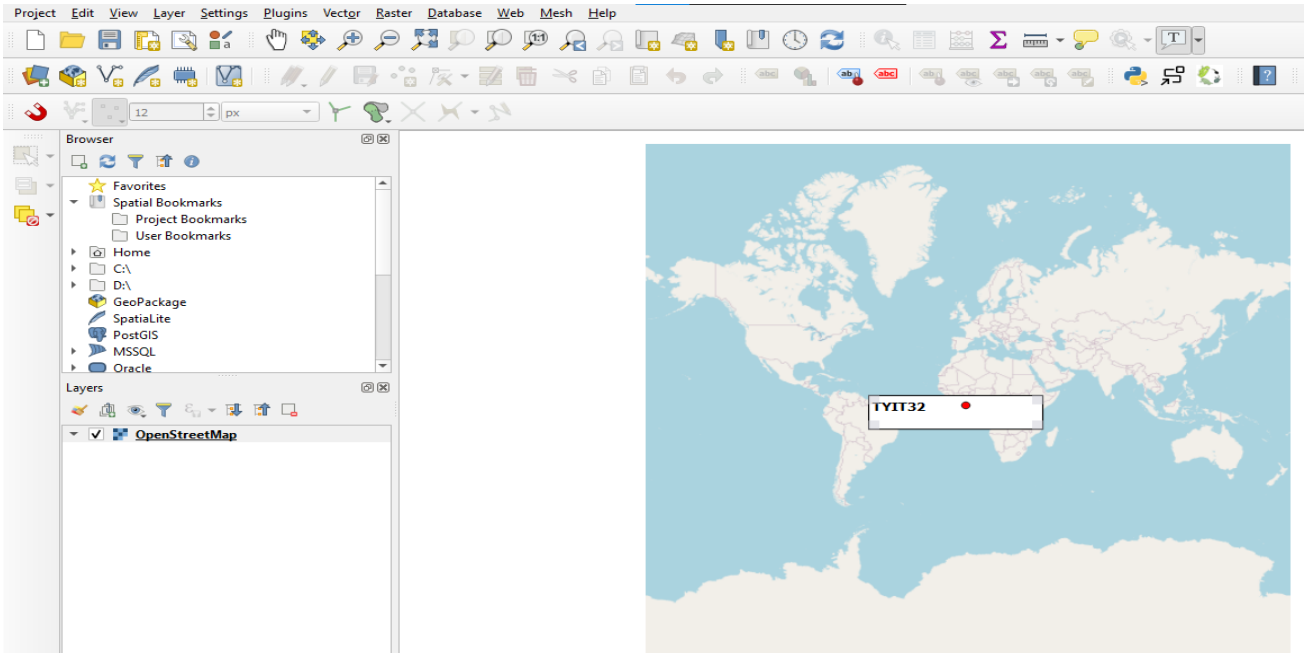
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

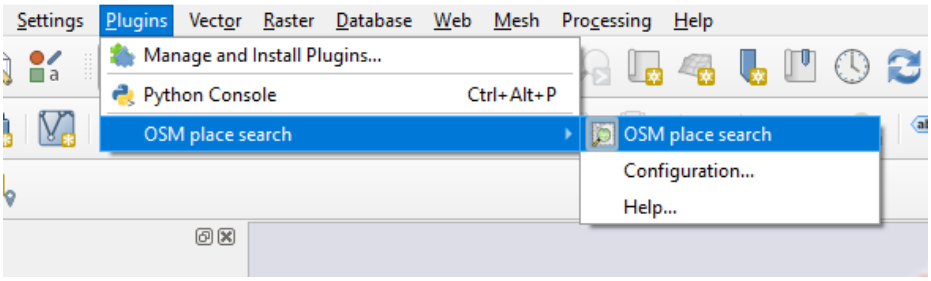
Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

Course Name: Geographic Information system **Page no:**

Practical Number:- 6



➢ Go to Plugins → select OSM Place search



➢ The Gateway of India, Mumbai is located at 18.92°N 72.83°E ➢ Search Gateway of India in OSM Search Pane

➢ Go to Raster → Georefrencer➢ File → Open Raster ➢ Select file "Gateway_Imagery.tif" from project

Teacher's Signature

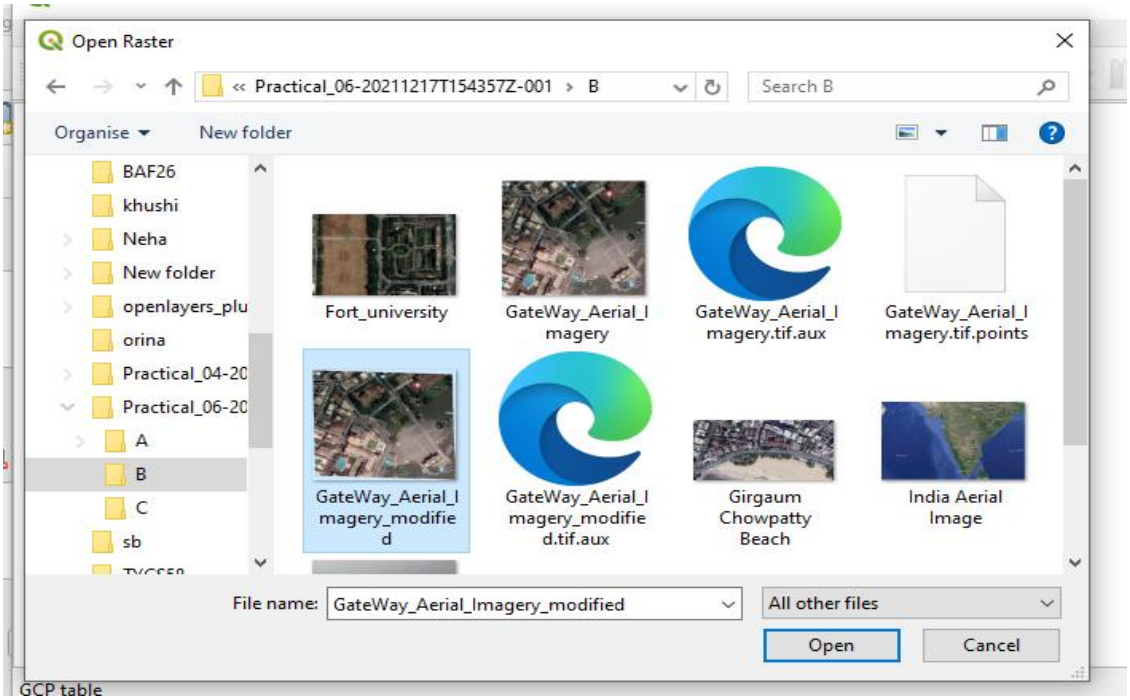
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

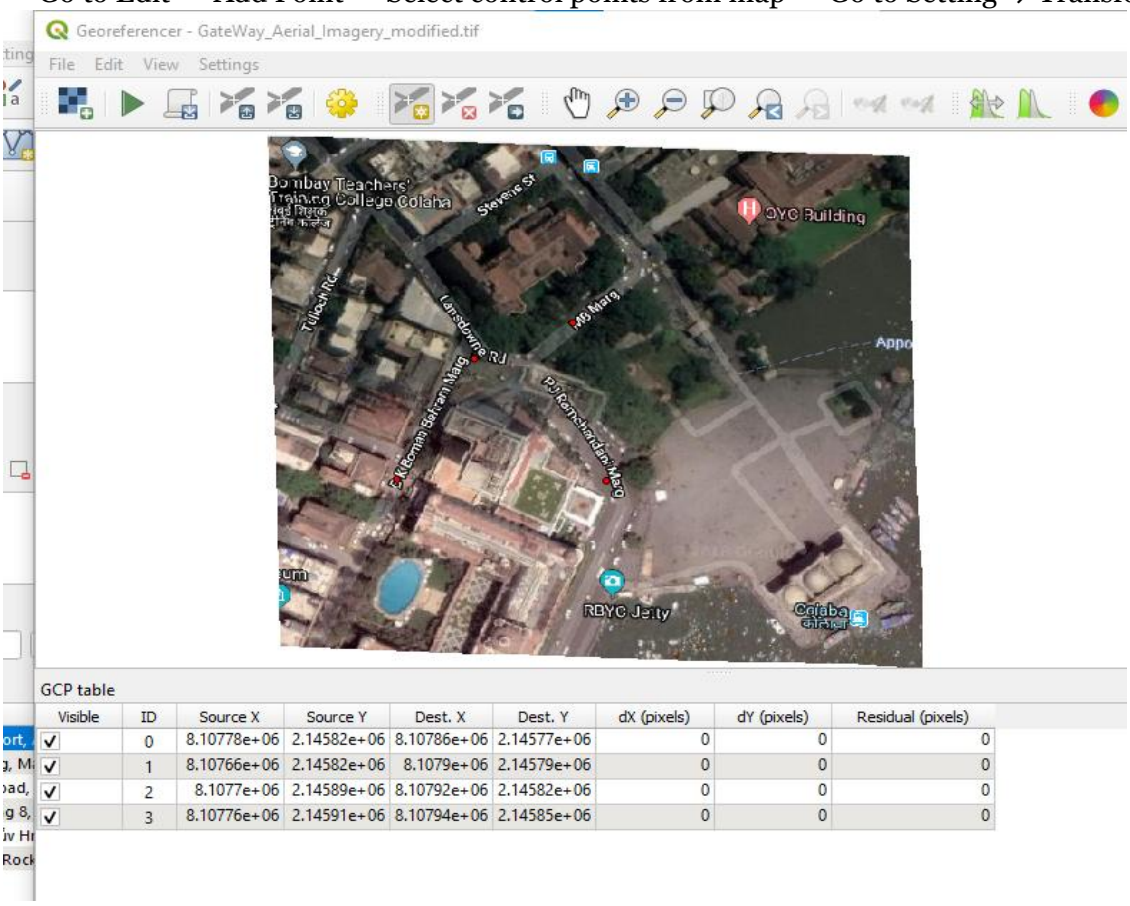
Course Name: Geographic Information system **Page no:**

Practical Number:- 6

data folder



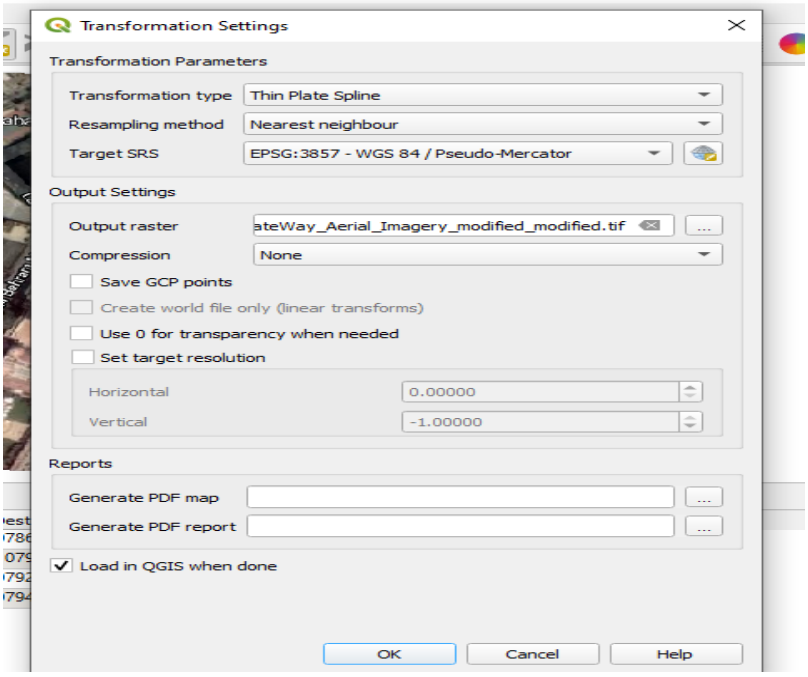
➢ Go to Edit → Add Point ➢ Select control points from map ➢ Go to Setting → Transformation Setting



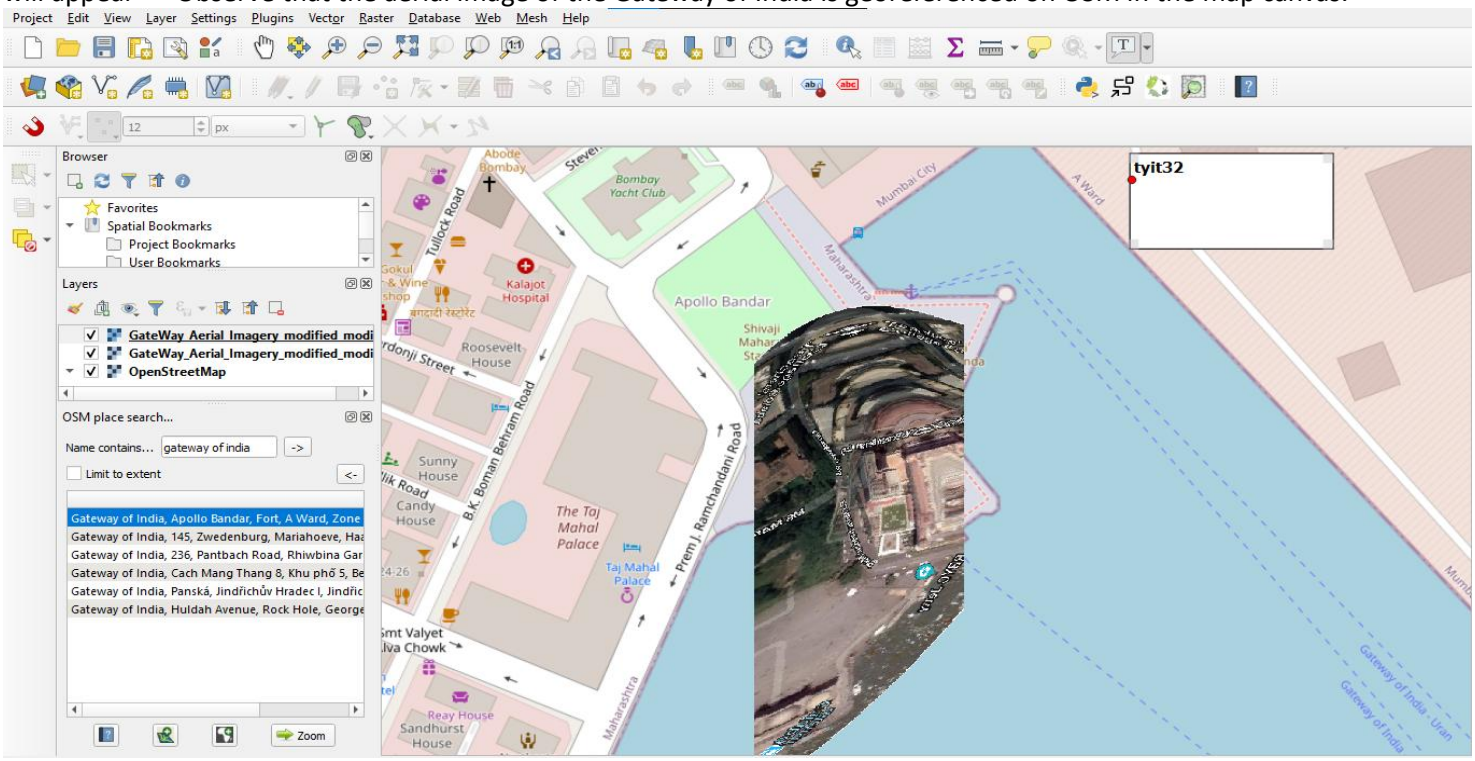
Visible	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	8.10778e+06	2.14582e+06	8.10786e+06	2.14577e+06	0	0	0
✓	1	8.10766e+06	2.14582e+06	8.1079e+06	2.14579e+06	0	0	0
✓	2	8.1077e+06	2.14589e+06	8.10792e+06	2.14582e+06	0	0	0
✓	3	8.10776e+06	2.14591e+06	8.10794e+06	2.14585e+06	0	0	0

Teacher's Signature

Practical Number:- 6



➢ Go to File → Start Georeferencing or Press the run button in Georegerencing Window. ➢ The progress indicator will appear ➢ Observe that the aerial image of the Gateway of India is georeferenced on OSM in the map canvas.



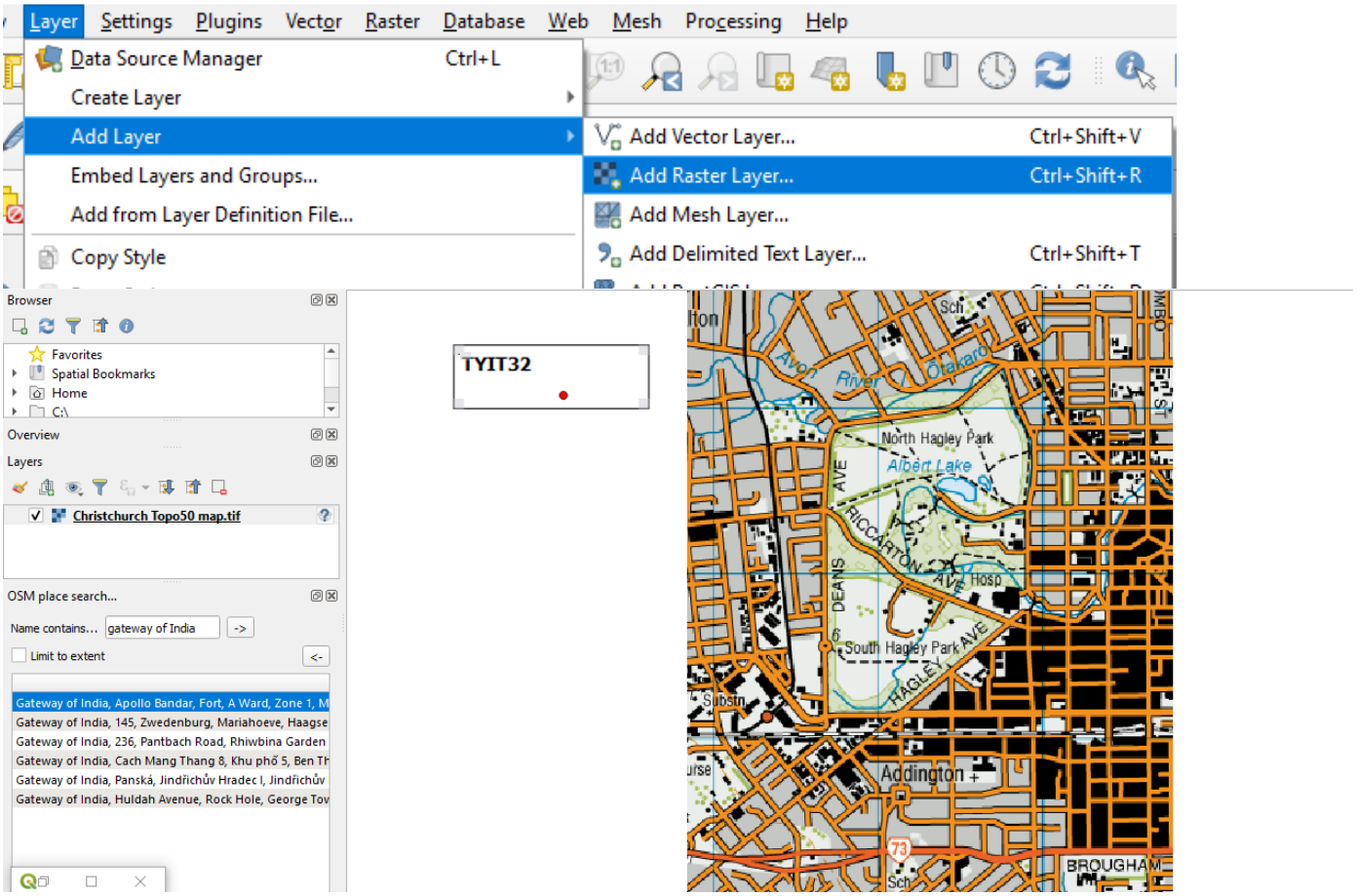
C. Digitizing Map Data

Spatialite is an open database format similar to ESRI's geodatabase format. Spatialite database is contained within a single file on your hard drive and can contain diferent types of spatial (point, line, polygon) as well as non-spatial layers. This makes is much easier to

Practical Number:- 6

move it around instead of a bunch of shapefiles.

Digitizing Map Data ➢ Go to Layer ‣ Add Raster→ Select "Christchurch Topo50 map.tif" from project Folder



➢ Right-click the Christchurch Topo50 map.tif layer and select Properties. ➢ Choose the Pyramids tab. Hold the Ctrl key and select all the resolutions offered in the Resolutions pane

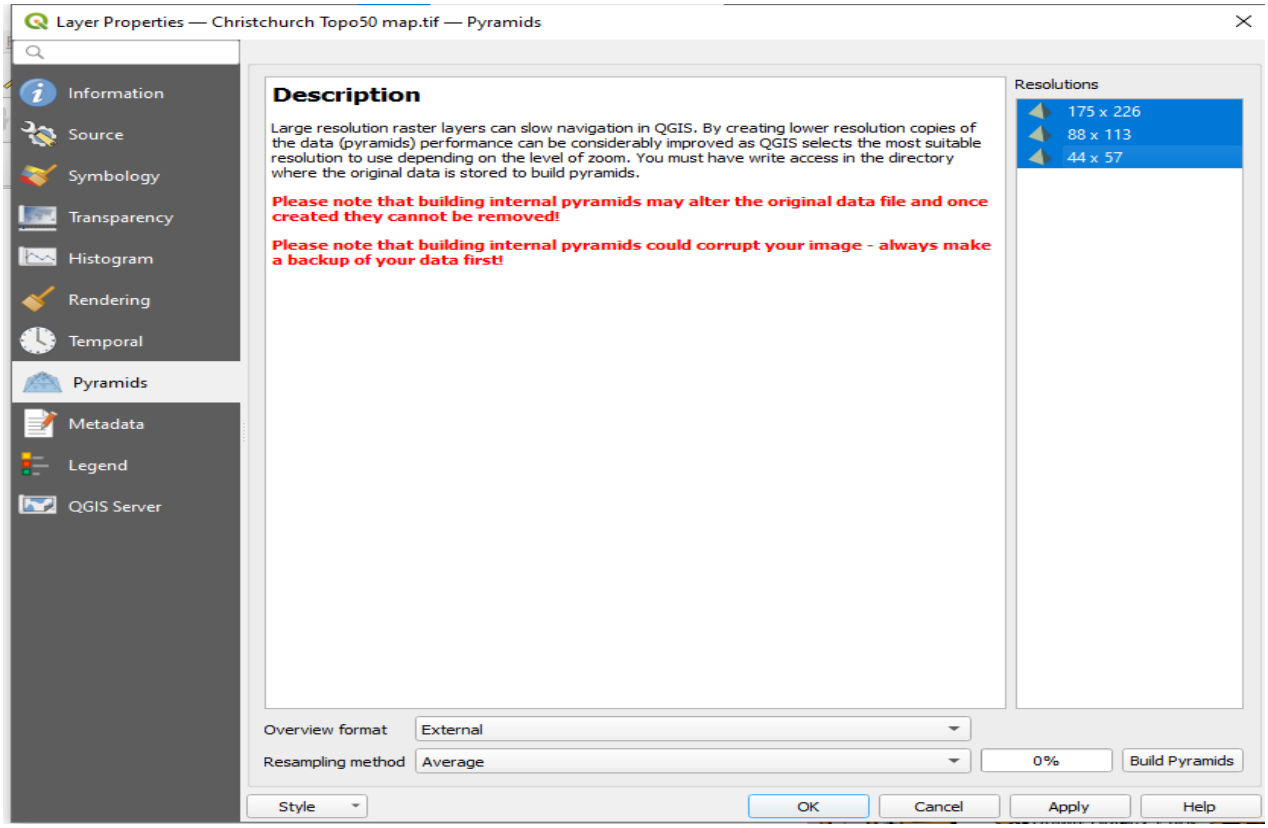
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

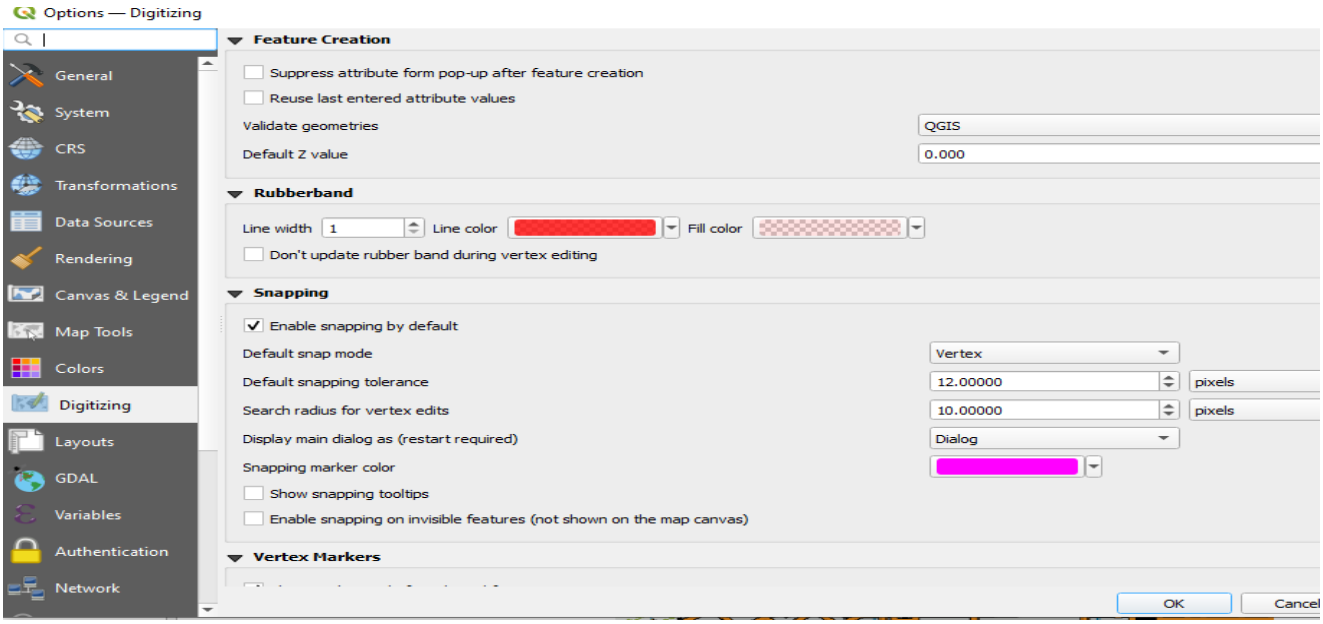
Course Name: Geographic Information system **Page no:**

Practical Number:- 6



➢ Click Build pyramids. Then click OK. ➢ Go to Settings →Options.... Select the Digitizing tab in the Options dialog.

➢ Set the Default snap mode to vertex and segment.



➢ Press OK.

➢ Go to Layer → Add Layer → Add Spatialite Layer

Teacher's Signature

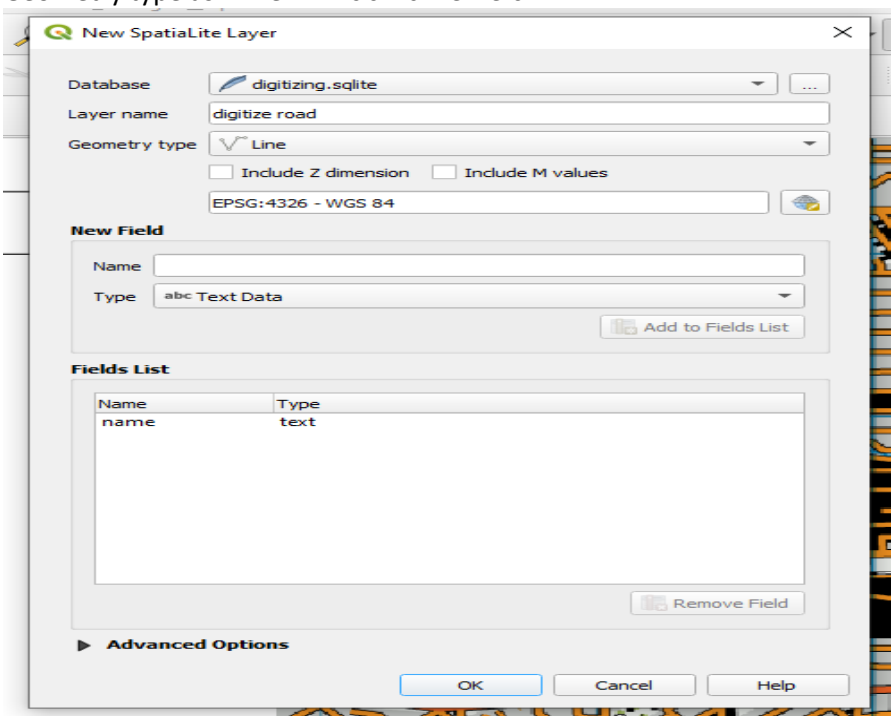
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

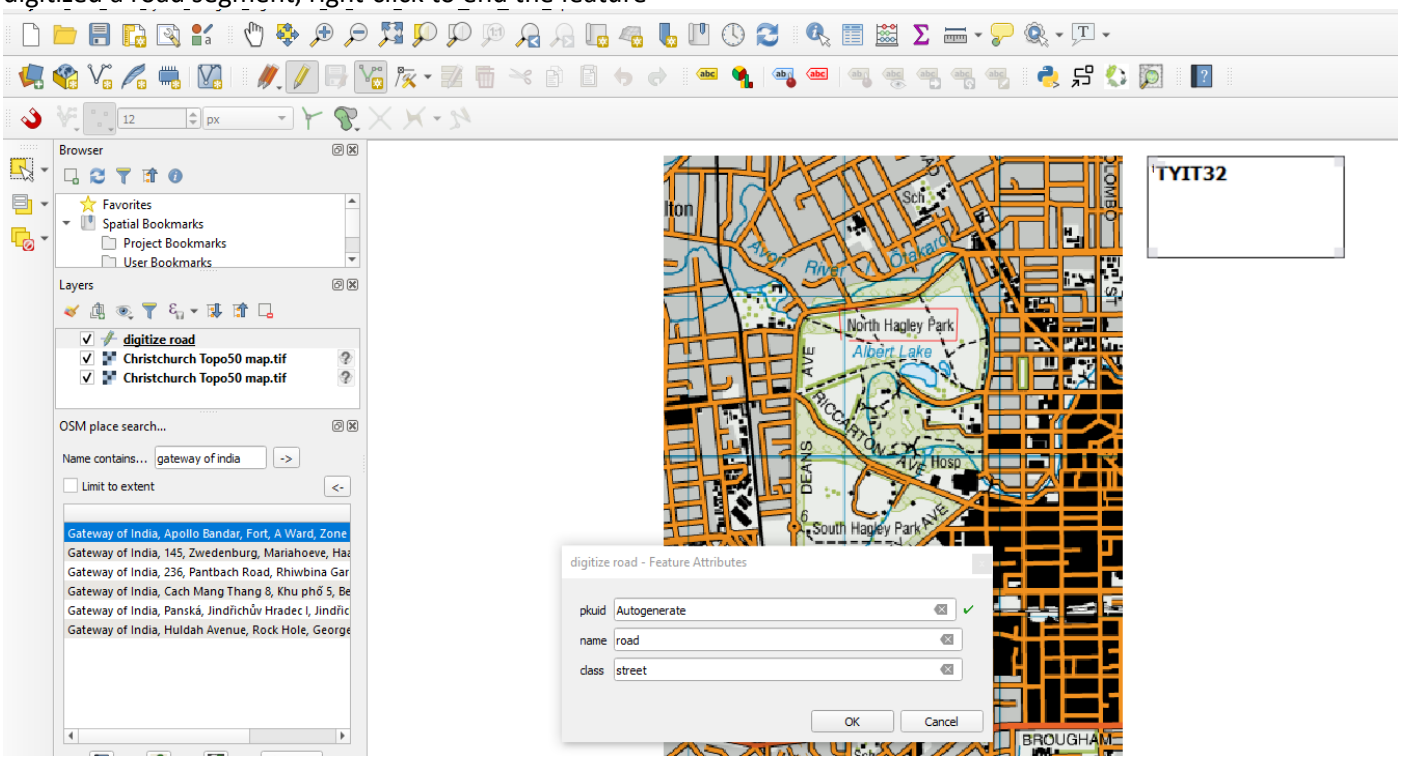
Course Name: Geographic Information system **Page no:**

Practical Number:- 6

➢ Select the name and location for Spatial database eg: "GIS_Workshop\Practicals\Practical_06\C\MySpatialDataBase.sqlite". ➢ Name the Layer as "Digitized_Road ➢ Set Geometry type as "Line" ➢Add name field



➢ Once the layer is loaded, click the Toggle Editing button to put the layer in editing mode. ➢ Click the Add feature button. Click on the map canvas to add a new vertex. Add new vertices along the road feature. Once you have digitized a road segment, right-click to end the feature



Teacher's Signature

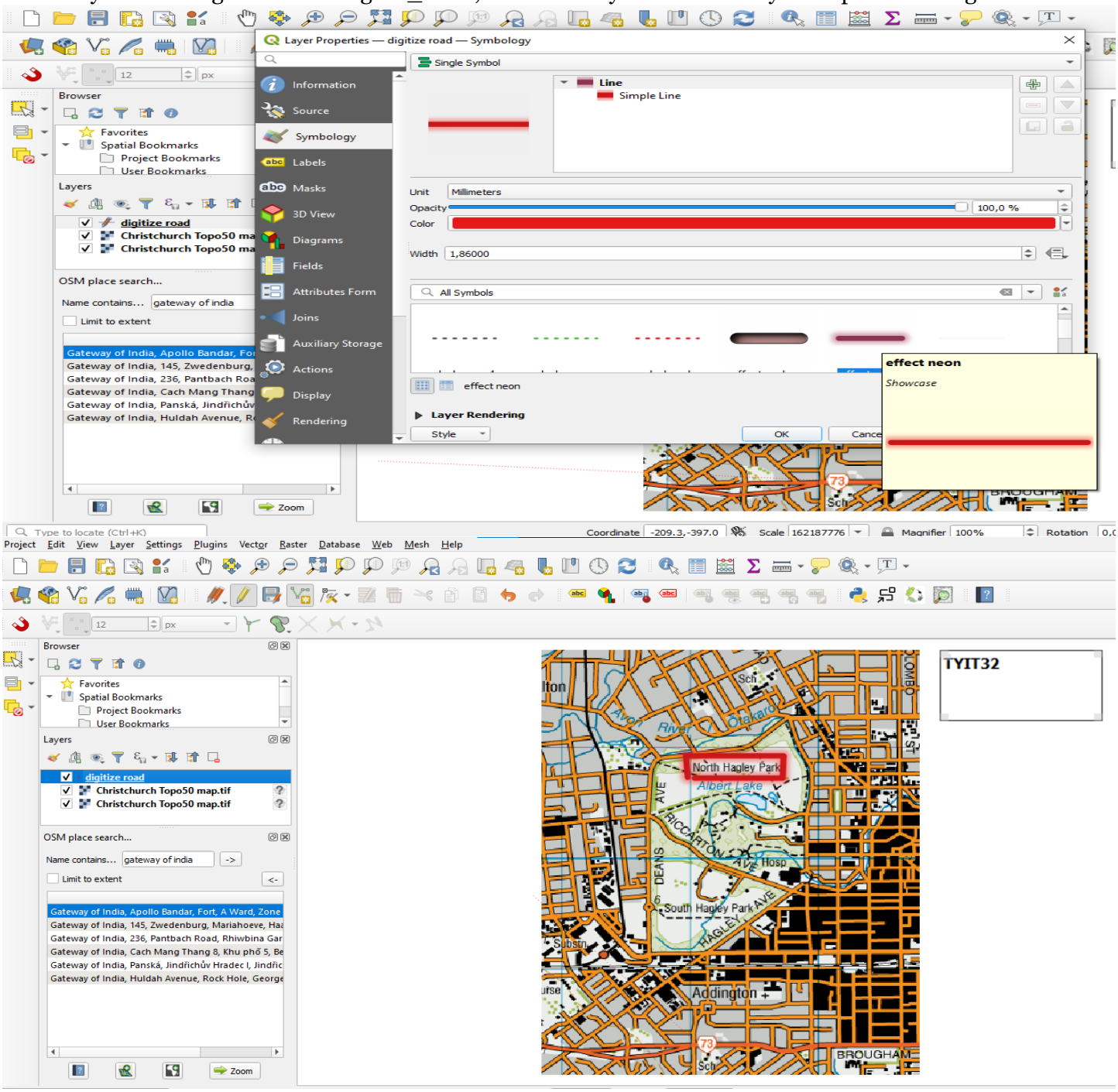
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

Course Name: Geographic Information system **Page no:**

Practical Number:- 6

➢ On Layer Panel Right Click on Digitze_Road, Select the Style tab in the Layer Properties dialog.



➢ After creating a new Spatialite layer

➢ Select building layer in Layer Panel and click on Toggle Editing button and then Add Polygon Feature button on Tool bar. ➢ Add two gardens to the region by adding polygon

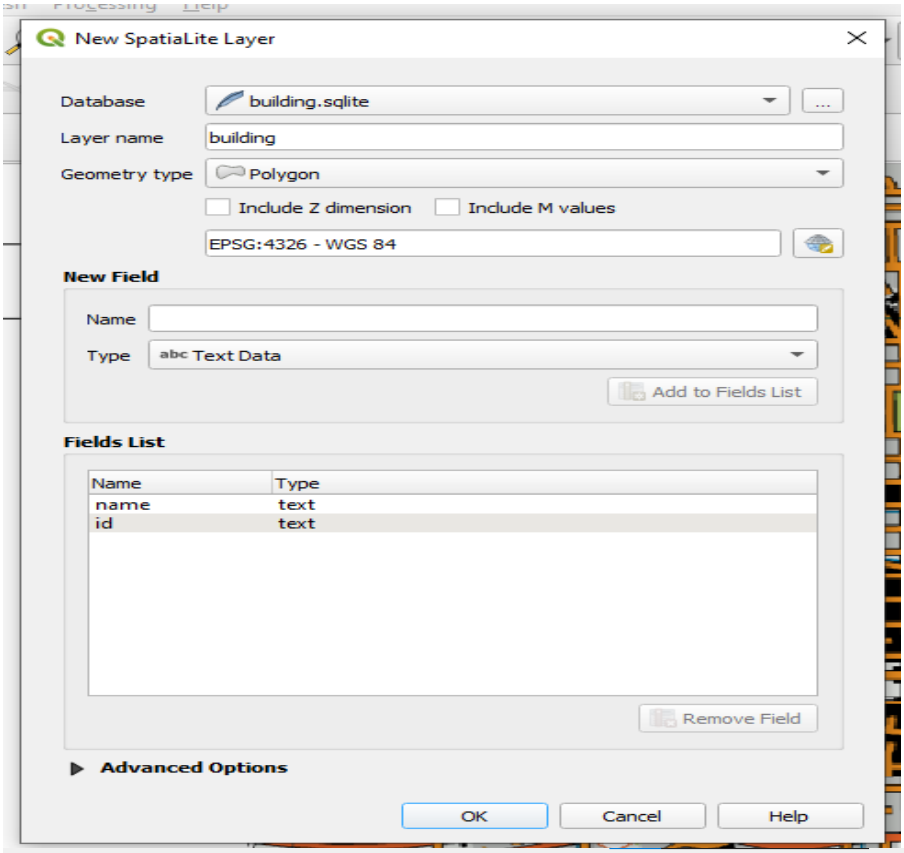
Teacher's Signature

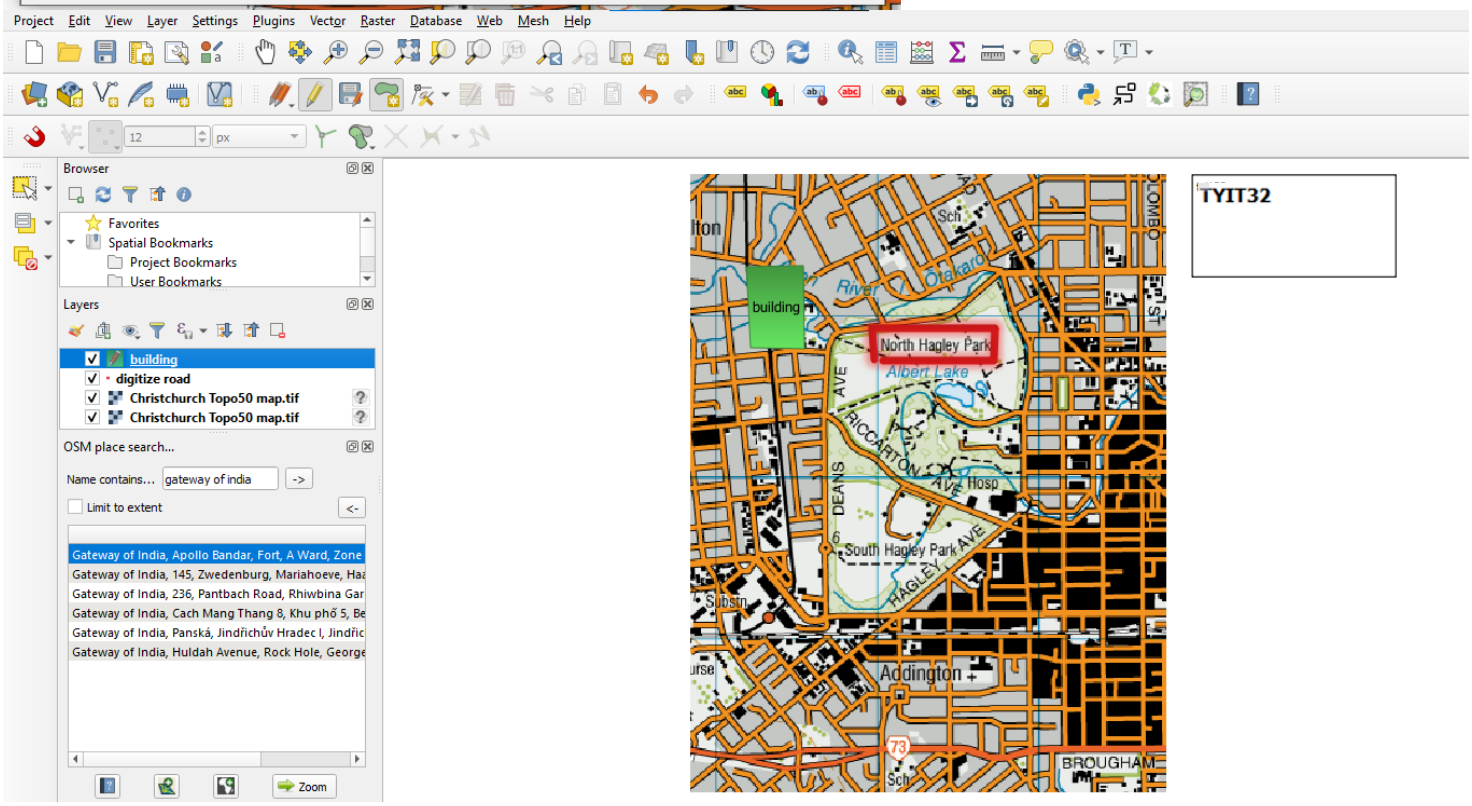
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT32** **Date: 20-01-2026**

Course Name: Geographic Information system **Page no:**

Practical Number:- 6





Teacher's Signature